

JALAY RUPERA

Backend Engineer | Distributed Systems | Python

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SUMMARY

Backend engineer with **4.5+ years** building distributed Python systems in production. Owns the asynchronous pipeline behind **344K+ patient insurance eligibility checks per week** (1.5M+ monthly) for **50+ hospitals and clinics across the UAE**, built on **FastAPI, Celery, Redis, RabbitMQ, and Kubernetes**. Founding-team and founder experience across two startups, with end-to-end ownership from REST API design through container orchestration, deployment, and production observability.

PROFESSIONAL EXPERIENCE

Software Development Engineer, Founding Team — Mantys

Apr 2025 – Present · Bangalore, KA

- Scaled the core insurance eligibility pipeline from **40K checks per month to 344K+ per week (~35x throughput)** through Celery queue partitioning, Redis caching, and worker concurrency tuning.
- Design and operate **LLM transformation workflows** on Azure OpenAI that convert unstructured HTML into structured JSON at **95%+ accuracy**, powering real-time eligibility verification across **50+ hospitals and clinics** in the UAE.
- Build automated data-retrieval pipelines in **Python and Playwright** against hospital portals and payer systems, cutting manual patient-data lookup time by **80%**.
- Ship and maintain the production Chrome extension used at hospital front desks — Emirates ID, Member ID, and DHA ID verification integrated with hospital information systems (HIS) for automated form filling and document upload — reducing user-reported issues by **90%**.
- Partner with the CTO on backend architecture, reliability, and observability for the core product, contributing to **\$2.5M+ in company ARR**.

Founder & Lead Backend Engineer — Vigilare Technologies

Jan 2024 – Feb 2025 · Bhavnagar, GJ

- Founded and led a **defense-tech startup** building Drishti, a **GPS-denied autonomous drone system** for mapping and inspection.
- Architected **FastAPI** backend services and REST APIs for **real-time telemetry streaming** and mission control between onboard compute and ground station.
- Built **ROS2** packages for **sensor fusion** and **Visual-Inertial Odometry (VIO)**, with MQTT-based telemetry transport across the drone-to-ground link.
- Containerized the full stack with **Docker** and deployed on **K3s**, enabling modular, independently versioned releases on edge hardware.

Backend Developer, B2B SaaS — Zu Technologies

May 2022 – Sep 2023 · Bangalore, KA

- Led the backend for Stock Wars, a real-time virtual stock-trading platform serving **10K+ concurrent users**, using Redis caching and message queues to hold latency stable under peak load.
- Built production **REST APIs and microservices** in Python (FastAPI) with MongoDB, designed for latency-critical trading workflows.
- Integrated **Celery, Redis, and RabbitMQ** for asynchronous job scheduling and background task execution.
- Deployed services on Docker and Kubernetes, improving deployment speed and reliability by **40%**.

Software Developer — Mindtree Limited

Jul 2021 – Feb 2022 · Bangalore, KA

- Built backend APIs and database schemas in **Python** for internal enterprise applications, and drove performance optimization and backend automation across client projects.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Bash

Backend: FastAPI, REST API design, microservices, asynchronous programming, system design

Distributed Systems: Celery, RabbitMQ, Redis, message queues, distributed task scheduling, caching, horizontal scaling

Data: MongoDB, Redis, schema design, data modeling

Infrastructure: Docker, Kubernetes (K8s/K3s), Azure, AWS, CI/CD, Linux, logging and observability

AI / LLM: Azure OpenAI, LLM pipeline orchestration, structured data extraction, prompt engineering

Tools: Playwright, Git, ROS2, MQTT

EDUCATION

B.Tech, Computer Science — REVA University, Bangalore

2017 – 2021